COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 2 : Session – 7 : SLO – 2 | BCD Addition
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Match the Following (Answers)

• 1. BCD → b. Binary representation of decimal digits
• 2. Correction rule → a. Add 0110 (6) if sum > 9
• 3. 1001 (BCD) + 0001 (BCD) → c. 1010 (Invalid BCD)
• 4. Valid BCD digit range → d. 0000 to 1001
• 5. Decimal 25 in BCD → e. 0010 0101

Fill in the Blanks

• BCD stands for BINARY CODED DECIMAL.
• In BCD, each decimal digit is represented by 4 bits.
• The valid range of a single BCD digit is 0000 to 1001 (0 to 9).
• If a BCD addition produces a sum greater than 1001, we must ADD 0110 (6) to correct it.
• The BCD sum of 0101 (5) and 0101 (5) after correction is 0001 0000 (decimal 10).

Activity – Describe and Perform BCD Addition (5 + 7)

• Steps: (1) Add the two 4-bit BCD digits as binary numbers. (2) If the 4-bit sum exceeds 1001 (9) or a carry is generated, the result is invalid BCD. (3) Add the correction factor 0110 (6) to that digit; the carry propagated to the next digit represents the tens digit.
• Perform: 0101 (5) + 0111 (7) = 1100 (12 in binary, invalid BCD).
• Correction: 1100 + 0110 = 0001 0010 → BCD digits 0001 (1) and 0010 (2) = decimal 12. Correct, since 5 + 7 = 12.